

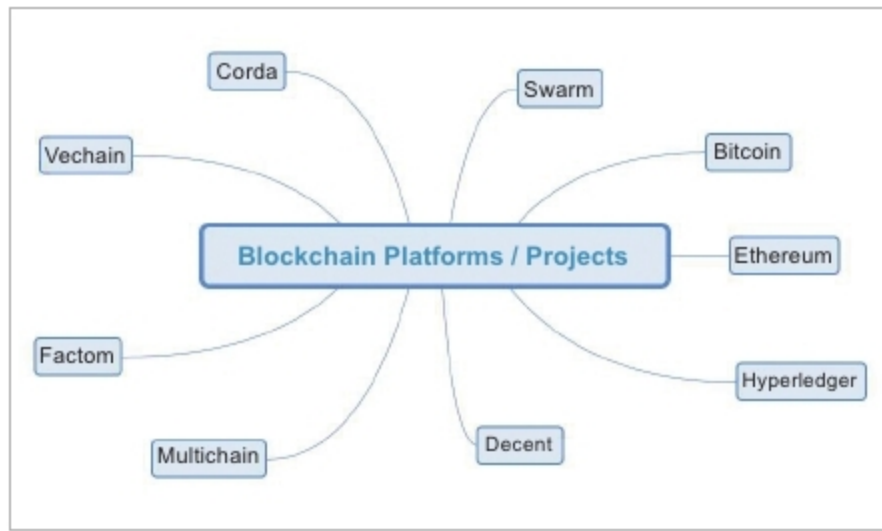


Figure 1: Blockchain projects and platforms

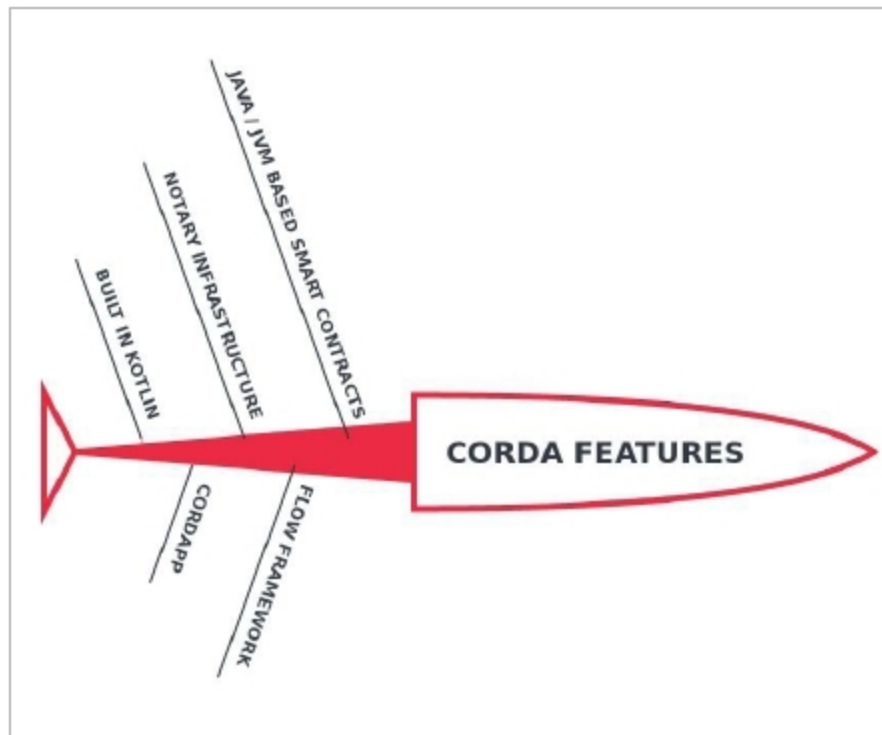


Figure 2: Corda's features

- States
- Transactions
- Contracts
- Flows
- Consensus
- Notaries
- Vault
- Time windows
- Oracles
- Nodes
- Transaction tear-offs
- Trade-offs
- Deterministic JVM

(Detailed instructions on each of these terms are available at <https://docs.corda.net/key-concepts.html>. Video based illustrations of these terms make it easier to understand them.)

The Corda network

Corda is a peer-to-peer network of nodes. Each node executes the Corda software and the Corda applications, which are termed CorDapps. The communication between these nodes is point-to-point and they are encrypted with the transport layer

security. The core point here is that data is shared only on a need-to-know basis, which removes the global broadcasts.

The ledger

In Corda, the data is not stored in a single centralised location. Each participating node holds its own database of known facts. This creates a situation where each peer only sees a subset of facts; no peer can be aware of the ledger in its entirety.

Corda DemoBench

The Corda DemoBench is a standalone desktop application. The DemoBench makes the configuration and launch of local Corda Nodes easy. Detailed instructions on DemoBench are available at <https://docs.corda.net/demobench.html>.

- Each node must have a unique name to identify it. The first node is the notary. So in this node, only the notary services are available. For the remaining nodes, other built-in services can be selected.
- Each new node is launched in the terminal emulator by DemoBench, which displays simple statistics about each node such as its cash balance.

The CorDapp

The CorDapps are Corda Distributed Applications. These apps run on the Corda platforms. The objective of the CorDapp is to allow nodes to reach an agreement on updates to the ledger. This is achieved by the flow definition, which the Corda node owners can invoke over RPC.

The CorDapp structure is similar to a set of JAR files containing class definitions coded in Java/Kotlin. The components of these class definitions are as follows.

- **Flows:** These define a routine for the node to run. They are usually for making updates to the ledger.
- **States:** These are used to define facts on which agreements are reached.
- **Contracts:** They constitute a valid ledger update.
- **Services:** These provide the long-lived utilities within the node.
- The Serialization Whitelist.

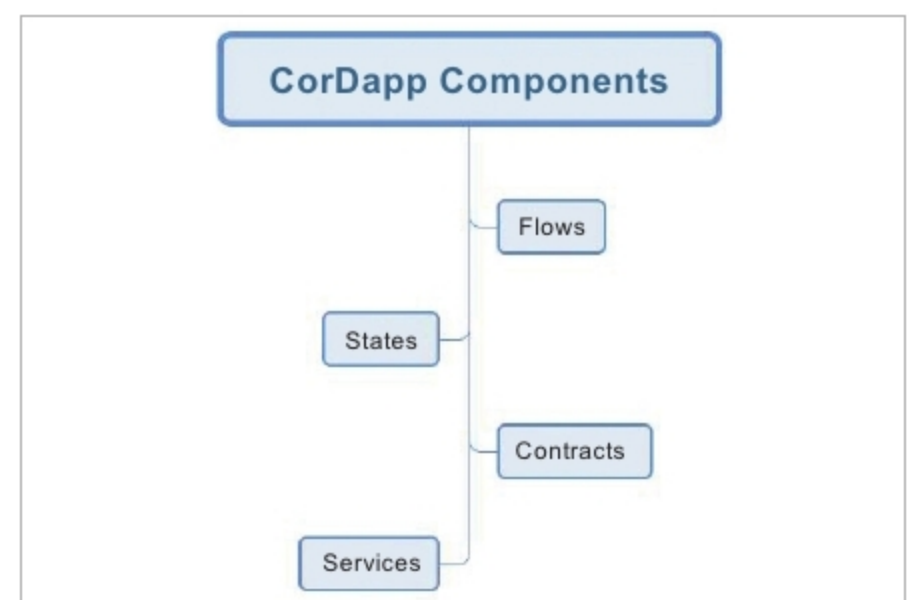


Figure 3: CorDapp components